

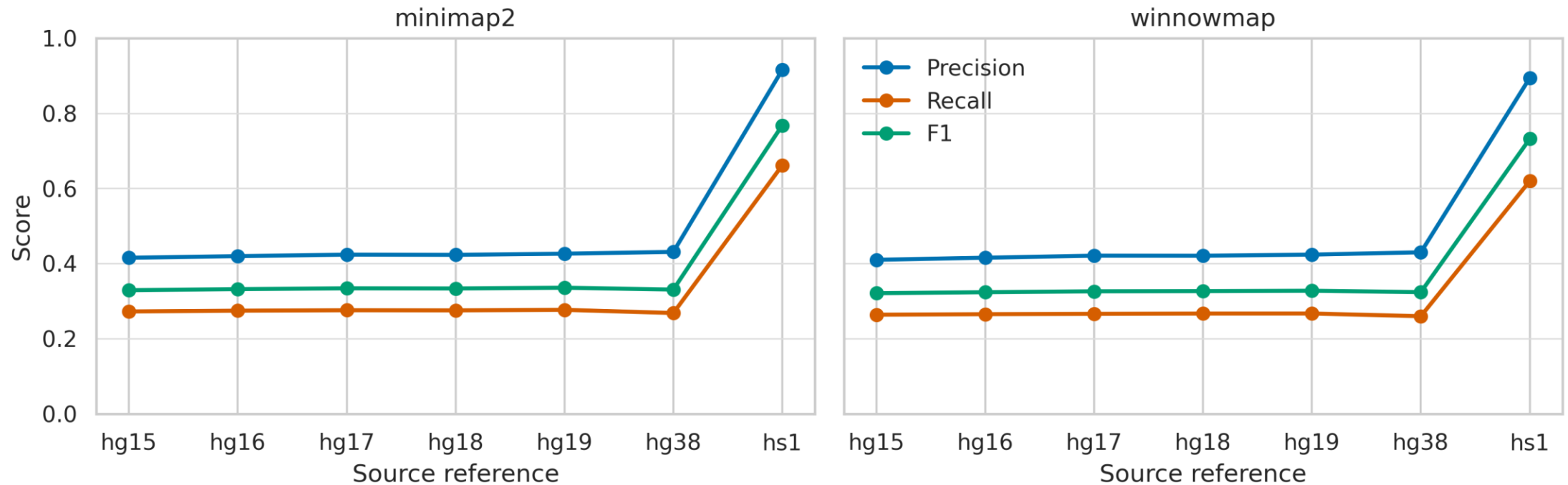
Benchmarking the impact of reference genome quality on SV discovery

HG002 PacBio HiFi, ~30× · seven reference builds, hg15 to T2T-CHM13 (hs1) · minimap2 and Winnowmap · Sniffles2 · GIAB v5.0q truth

Group 4 - “SVCrying” · Results

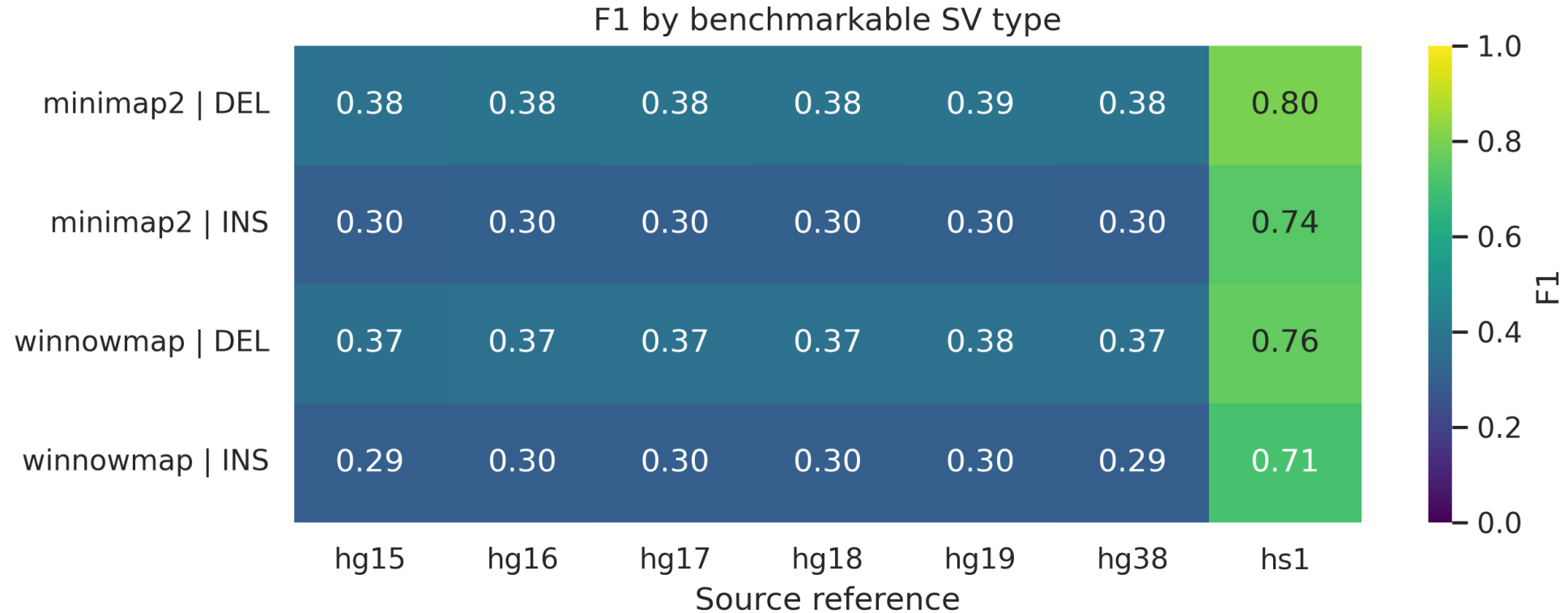
Reference choice matters far more than mapper choice

HG002 SV benchmark against GIAB hs1 truth



- Native hs1 calling reaches F1 ≈ 0.77 (minimap2); every lifted callset sits near 0.33.
- Older builds are flat — no consistent gain from hg15 through hg38. minimap2 only modestly ahead of Winnowmap.

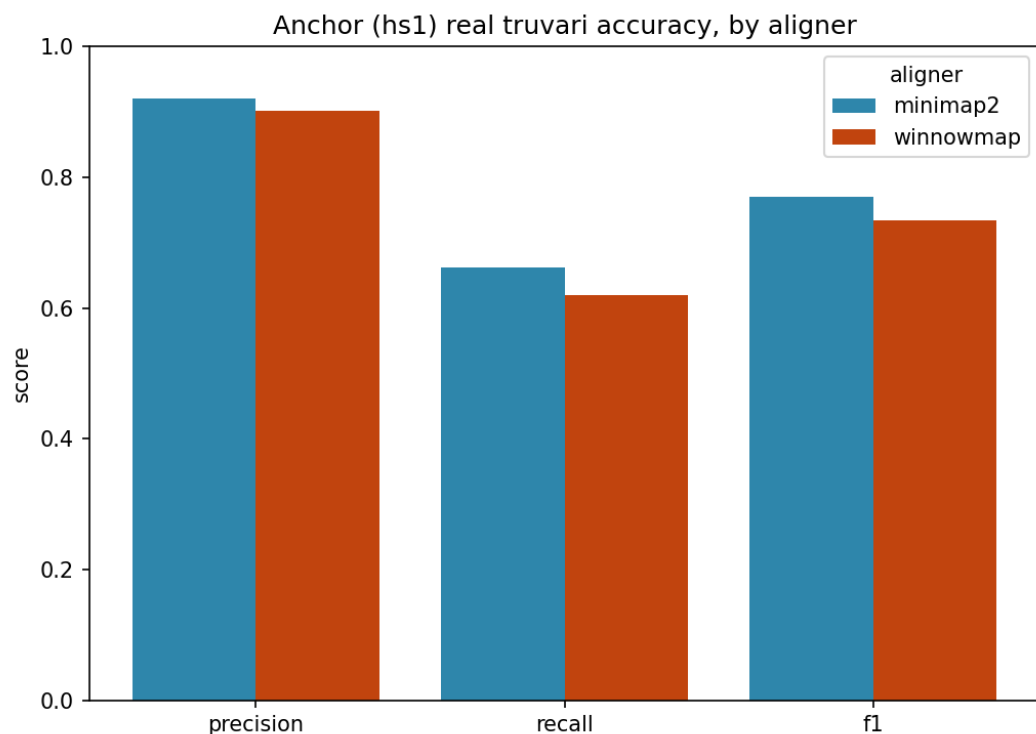
Deletions are recovered more reliably than insertions



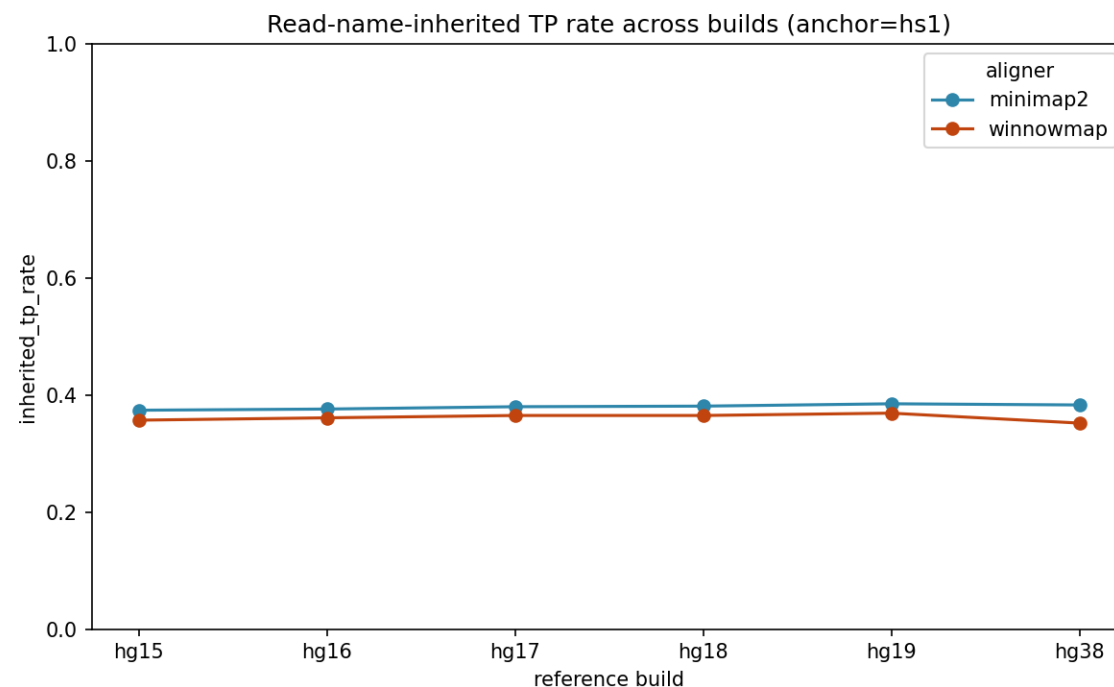
- Insertions sit roughly 0.07–0.09 F1 below deletions in every reference and both aligners.
- Performance also declined with SV length, with very large events recovered least reliably.

Without coordinate conversion, the reference ladder flattens

Only hs1 is scored against truth. Other builds are matched by supporting-read overlap with hs1's confirmed true positives — an inherited TP rate, not a recall.



Anchor (hs1) vs GIAB v5.0q — P/R/F1 0.92/0.66/0.77 (minimap2), 0.90/0.62/0.73 (Winnowmap)



Inherited TP rate flat across hg15–hg38: 0.374–0.385 (minimap2), 0.352–0.369 (Winnowmap)

The two benchmarks disagree, and that is the finding

Liftover benchmark

- hs1 wins on two counts at once: better sequence representation and no coordinate projection.
- The two cannot be separated in this design.

Read-name benchmark

- Flat across builds, so reference quality matters much less once liftover is removed.
- But 55–59% of calls share no reads with an anchor TP and are unclassified, not false; 6–7% are dropped on SVTYPE mismatch.

What drives reproducibility

- SVs concentrate in simple repeats, satellites, segmental duplications and low-mappability sequence — for TPs, FPs and FNs alike.
- Clusters are mostly all-or-nothing: shared across nearly all builds, or confined to one.

The strong hs1 result should be read as improved representation plus the absence of liftover and truth-coordinate mismatch — not as superior mapping performance.